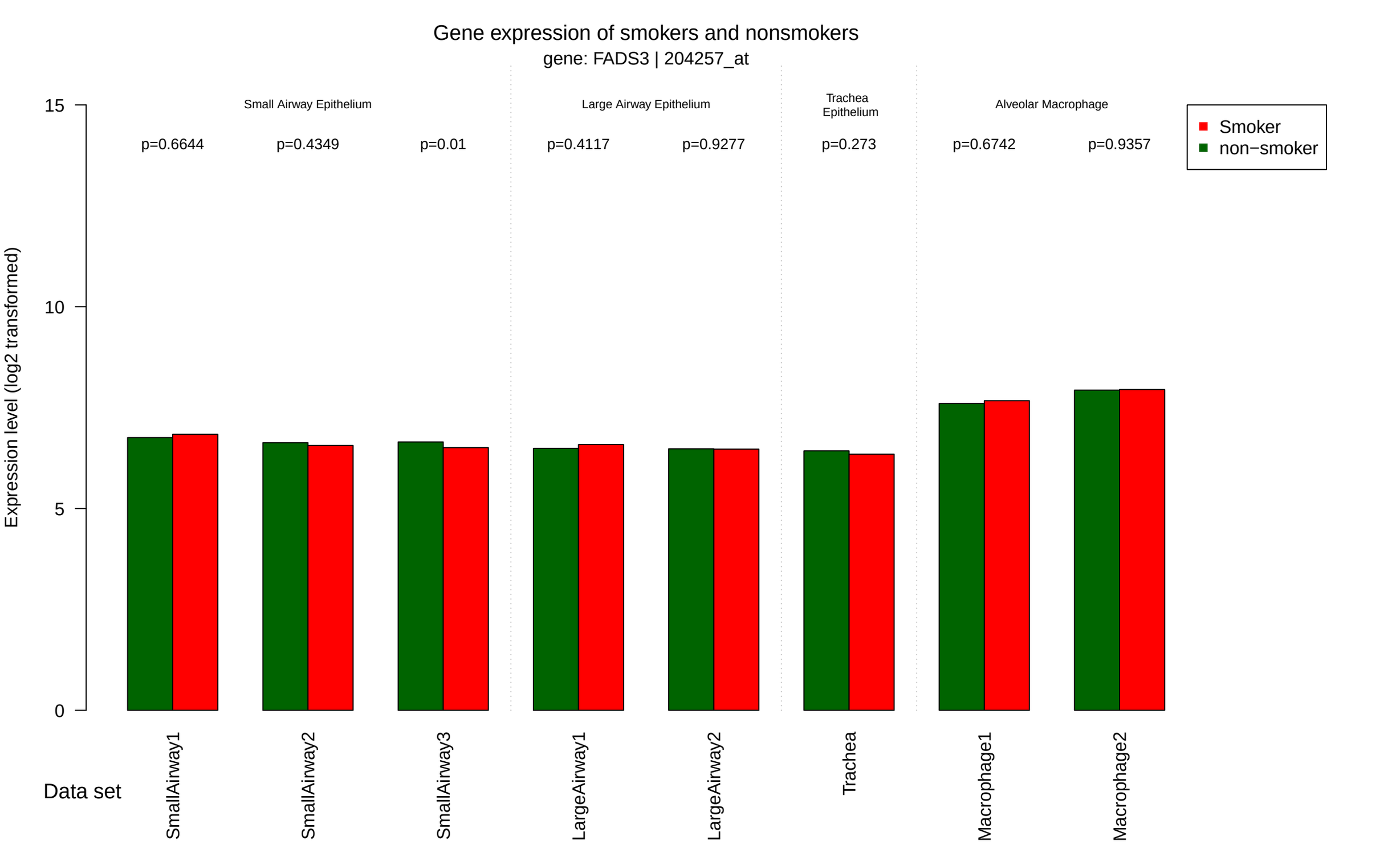
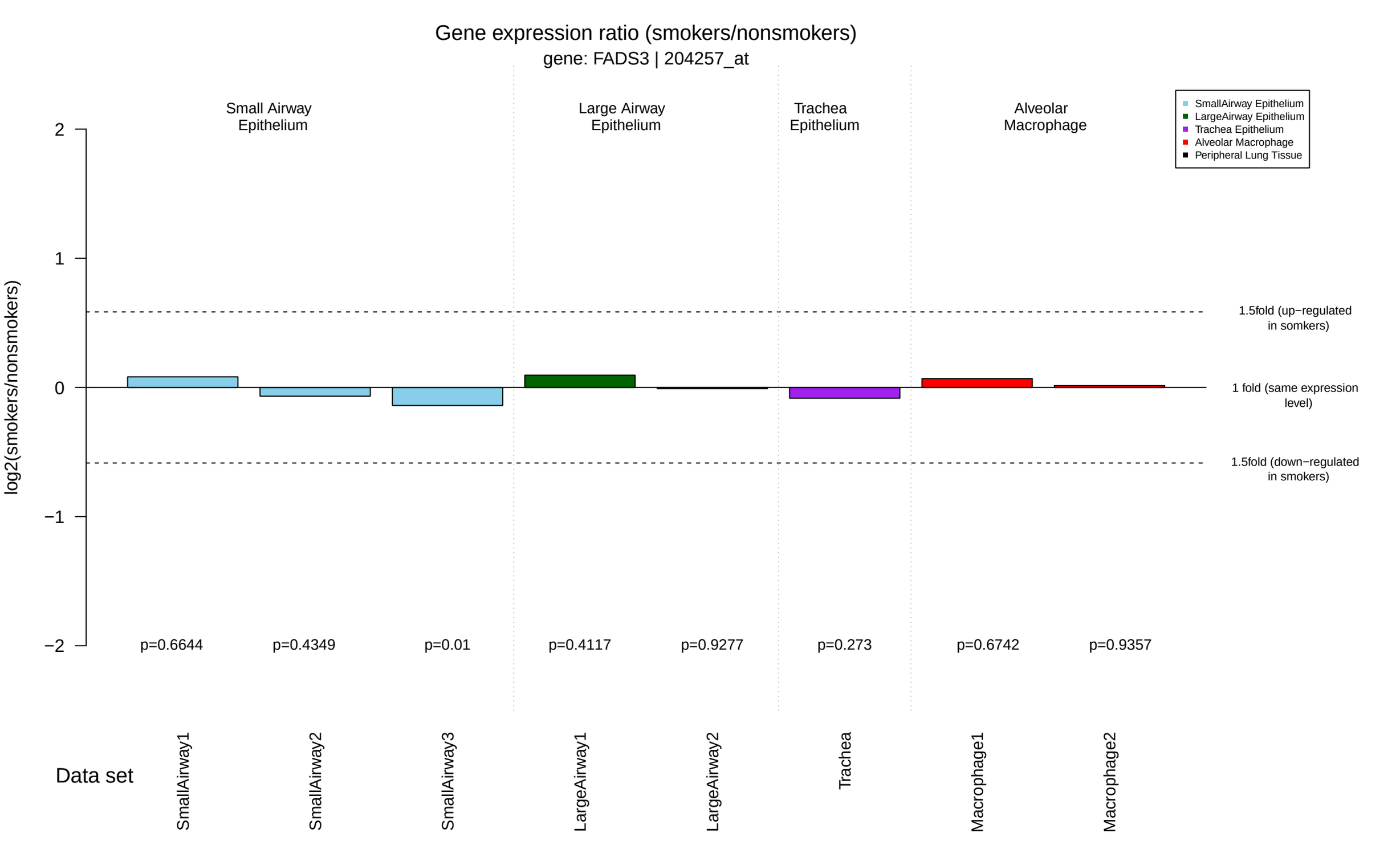
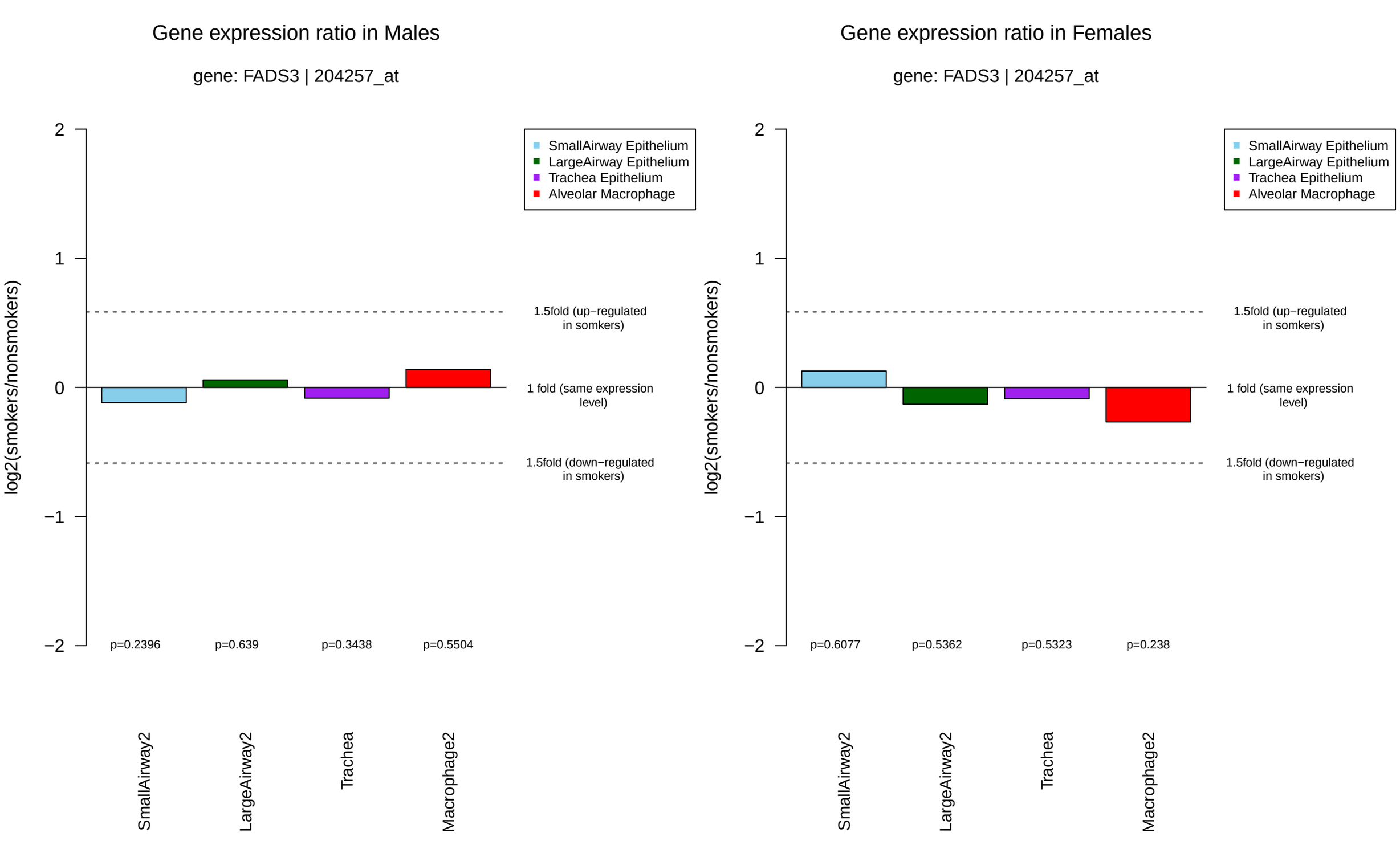
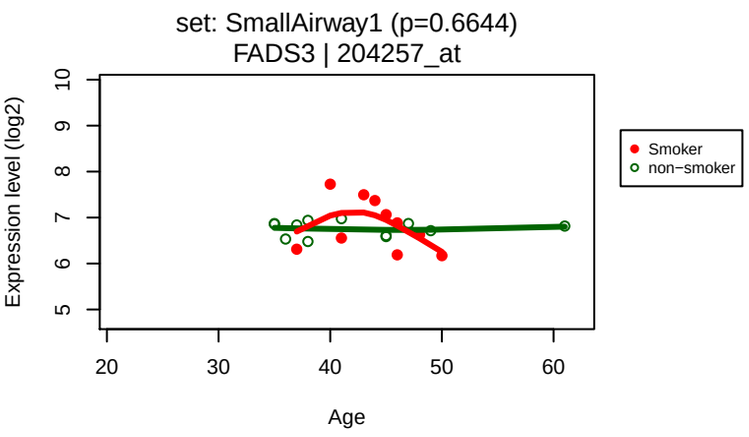
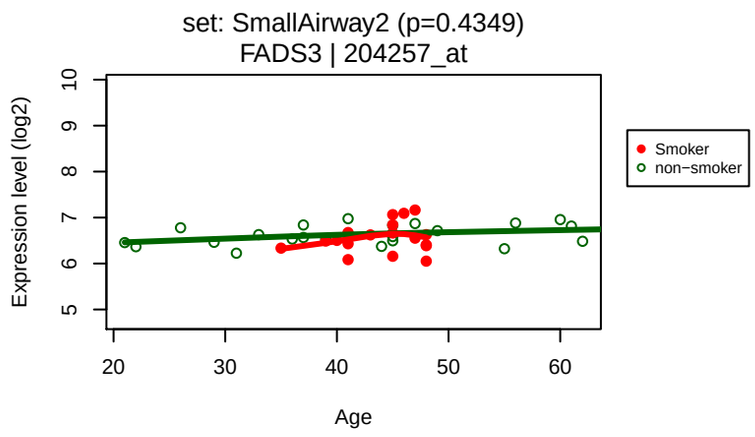
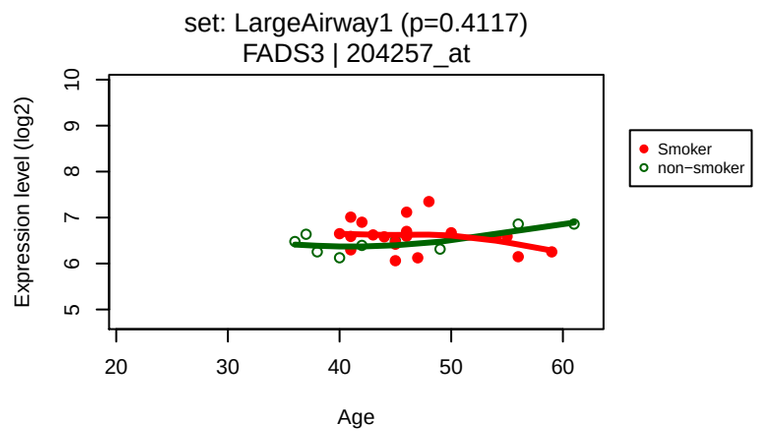

Gene expression of smokers and nonsmokers
gene: FADS3 | 204257_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6644
p=0.4349
p=0.01
p=0.4117
p=0.9277
p=0.273
p=0.6742
p=0.9357
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

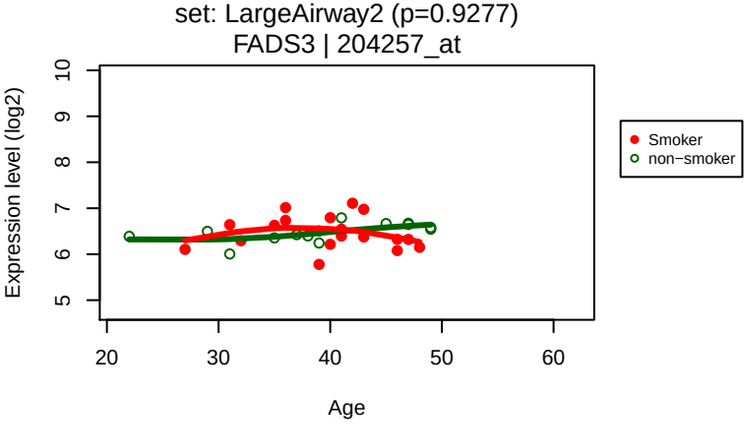

Gene expression ratio (smokers/nonsmokers)
gene: FADS3 | 204257_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6644
p=0.4349
p=0.01
p=0.4117
p=0.9277
p=0.273
p=0.6742
p=0.9357
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

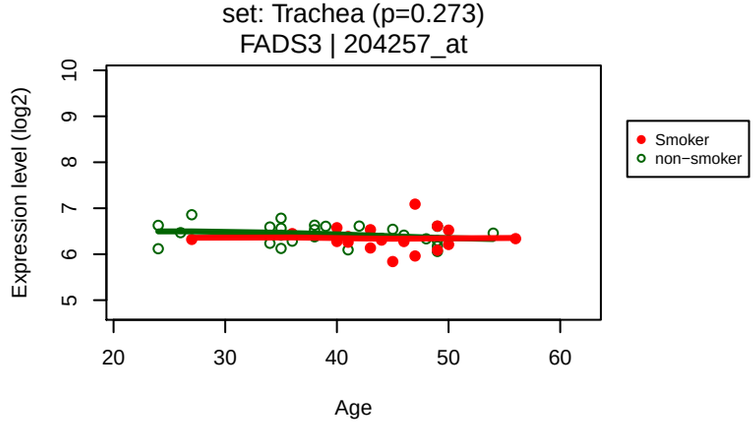

Gene expression ratio in Males
gene: FADS3 | 204257_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2396
p=0.639
p=0.3438
p=0.5504
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: FADS3 | 204257_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6077
p=0.5362
p=0.5323
p=0.238
SmallAirway2
LargeAirway2
Trachea
Macrophage2

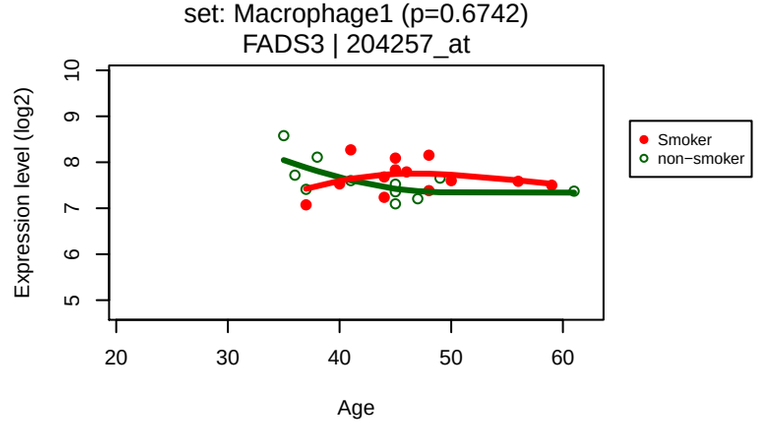

set: SmallAirway1 (p=0.6644)
FADS3 | 204257_at
Expression level (log2)
Age
Smoker
non−smoker

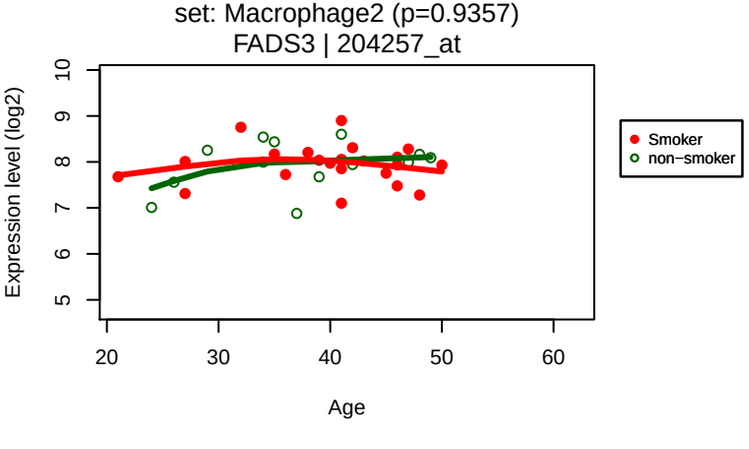

set: SmallAirway2 (p=0.4349)
FADS3 | 204257_at
Expression level (log2)
Age
Smoker
non−smoker

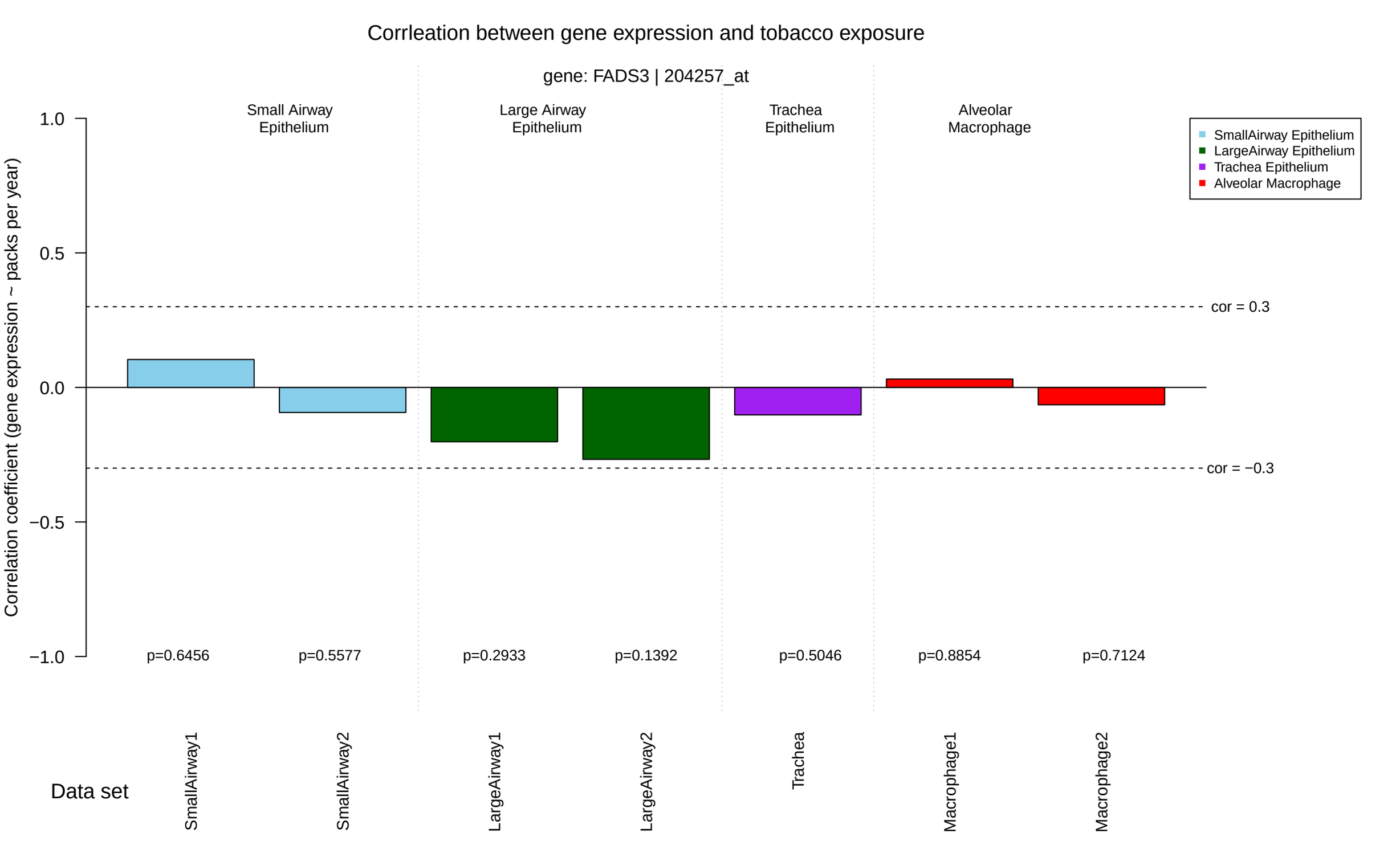

set: LargeAirway1 (p=0.4117)
FADS3 | 204257_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9277)
FADS3 | 204257_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.273)
FADS3 | 204257_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6742)
FADS3 | 204257_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9357)
FADS3 | 204257_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: FADS3 | 204257_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6456
p=0.5577
p=0.2933
p=0.1392
p=0.5046
p=0.8854
p=0.7124
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2